



alex bellon ('balex')

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I like to take things apart and understand how they work. My experience spans reverse engineering, security analysis, embedded firmware development (especially involving WiFi/BLE), and trawling through specifications, datasheets, patents, obscure documentation, FCC filings, and more.

EDUCATION

- 2021–Present **Ph.D., Computer Science, University of California San Diego**
Advised by [Deian Stefan](#) and [Pat Pannuto](#), with a focus on the security of high-risk embedded devices.
[SysNet](#) and [CryptoSec](#) groups.
- 2017–2021 **B.S., Computer Science, The University of Texas at Austin**
- 2017–2021 **B.S., Mathematics, The University of Texas at Austin**

SELECTED PUBLICATIONS * = first (co)author

- [4] **The Case for Crowd-Sourcing Communication to BLE-Only Devices**
Alex Bellon*, Tess Despres*, Prabal Dutta, Pat Pannuto, Sylvia Ratnasamy
24th ACM Workshop on Hot Topics in Networks (HotNets 2025)
- Coauthored proposal to secure \$100,000 Qualcomm Innovation Fellowship
 - Designed protocol allowing direct downlink communication to BLE devices using nearby smartphones
 - Implemented proof-of-concept system using ESP32, Python and C++
 - Performed measurements showing ephemeral, opportunistic BLE connections are long enough to support 100kB data transfers (devices in this class often have <1MB flash)
- [3] **Hacking for Fun and Glucose: Reverse Engineering an Insulin Pump**
Alex Bellon*, Alex Snoeren, Deian Stefan
SRC TECHCON 2022
- Disassembled and reverse-engineered the hardware of an STM32/nRF52-based insulin pump
 - Developed custom PCB to connect to proprietary debug connectors and allow for firmware to be extracted over SWD and JTAG using STLinkV2 and JLink programmers
 - Identified interesting functions in firmware using Ghidra
 - Worked with manufacturer to responsibly disclose security vulnerabilities found in firmware
- [2] **TagAlong: A Free, Wide-Area Data-Muling Service Built on the AirTag Protocol**
Alex Bellon*, Alex Yen, Pat Pannuto
24th International Workshop on Mobile Computing Systems and Applications (HotMobile 2023)
- Led development of data exfiltration system using the AirTag protocol and existing infrastructure
 - Developed a proof-of-concept system with ESP32s, C++, Python, and Swift to embed arbitrary data in existing AirTag FindMy packets
- [1] **An Empirical Analysis on the Use and Reporting of National Security Letters**
Alex Bellon, Miro Haller*, Andrey Labunets, Enze Liu, Stefan Savage
4th ACM Symposium on Computer Science and Law (CSLAW 2025)
- Manually located and evaluated corporate transparency reports from 50+ companies over a span of 15 years to determine the amount of NSL requests they received
- [0] **Understanding the viability of e-mail origin indicators for identifying the sender**
Enze Liu*, Lu Sun, **Alex Bellon**, Grant Ho, Stefan Savage, Geoffrey M. Voelker, Imani N. S. Munyaka
19th USENIX Symposium on Usable Privacy and Security (SOUPS 2023)
- Assisted in the design of IRB-approved user study to evaluate the effectiveness of security UI features

RESEARCH EXPERIENCE

- Summer 2026–Present **Visiting PhD Student, UC Berkeley (with [Prabal Dutta](#) and [Lab11](#))**
- **Privacy-first downlink communication with BLE-only edge devices**
 - Designed network for direct downlink communication to BLE-only devices using opportunistic smart-phone connections as backhaul infrastructure
 - Led threat modeling and selection of design protections like AEADs (ChaCha20-Poly1305), key exchange/derivation (Diffie-Hellman, BLAKE3)
 - Implemented protocol for servers and BLE edge devices using Rust and C/C++
 - Created simulator in Python to evaluate system performance using real BLE trace captures
 - Evaluated system's end to end latency (400KiB in <1min. on avg.) and latency overhead (min. 7.6%)

- Summer 2021–Present **Graduate Student Researcher, UC San Diego**
- **Evaluating the security of WiFi and BLE infrastructure using weak authentication handshakes**
 - Discovered “quirk” in WPA/WPA2-PSK and BLE authentication specifications that allows passwords to be tested without fully authenticating to the device (thereby not violating access laws)
 - Developed tool using Python, scapy, bumble, bash and C to confirm a device is using a given password
 - Reverse engineered various access point firmware and Android applications using JADX, Ghidra, Wireshark, to determine the authentication method used and what logging is enabled
 - Discovered vulnerability in common AP driver that allows for multiple password guesses per session
 - Used custom tool to evaluate default credential usage against real-world devices, capturing interactions using Wireshark, nRF BLE sniffer, Android HCI logging
 - **Security analysis of a commercial IFE/in-flight WiFi system**
 - Extracted and rehosted encrypted Linux filesystem from commercial IFE hardware, cracked password
 - Reverse engineered configuration files and programs to map network topology, identify attack vectors
 - Crafted attacks to change networking configurations from passenger network
 - **Firmware-in-the-loop verification of an insulin pump**
 - Assisted in extracting execution logic from binary ARM-based firmware for commercial insulin pump and rehosting relevant portions using Python and Unicorn to execute as needed
 - Implemented model of human pancreatic system to interact with insulin pump emulator and perform closed loop simulation for verification purposes using Python, Verse, and SciPy
 - Used standard medical classification scale for blood glucose levels to determine “safety” rating of insulin pump’s behaviors given an initial set of parameters
 - **Additional side projects**
 - Added support for Motorola 68000 architecture to emulation tool
 - Developed firmware for ESP32 and HelTec boards to broadcast measurement packets over LoRa and BLE and record packets locally on flash
 - Developed custom PCB with LED, BLE, heart rate sensing functionality using KiCAD, implemented and debugged firmware using C, STLinkV2, logic analyzer
- Fall 2023 **Research Intern, Max Planck Institute for Security and Privacy, Bochum, Germany**
- **Evaluating the security of an Android insulin pump simulator**
 - Extracted insulin pump simulation libraries from demo Android application, reverse engineered using JADX, JD-GUI, Frida and Ghidra
 - Created a custom harness using Android NDK, C++, and Android emulators to rehost simulation libraries and connect to fuzzer

INDUSTRY EXPERIENCE

- Summer 2023 **Software Engineering Intern, Micron, San Jose, CA**
- Created optimization passes for LLVM/Clang to improve workload performance with CXL memory
 - Added pragma to allow for replacement of more CXL-efficient instructions in C++ programs
- Summer 2020 **Security Engineering Intern, Mozilla, Mountain View, CA (remote)**
- Researched security issues in language-based package managers like Cargo, NPM and PyPI
 - Used prior work to fix security scoring algorithm on Mozilla’s Dependency Observatory (github.com/mozilla-services/dependency-observatory) project, used to “rate” the security of NPM packages
- Summer 2019 **Security Analyst Intern, Electronic Arts, Seattle, WA**
- Used Python to automate checking for open ports and other attack vectors on EA’s cloud instances. Scanned 800+ instances, found 1400+ security incidents
 - Automated sending summary of vulnerabilities to affected parties, with descriptions of the vulnerabilities and instructions to resolve them

TEACHING EXPERIENCE

- Spring 2026 **Teaching Assistant, CSE107 Cryptography, UC San Diego**
- Led weekly discussion sections, review sessions near exam time
 - Graded assignments, tests, and held office hours for a class of 50 students
- Spring 2021 **Undergraduate TA, CS349 Contemporary Issues in Computer Science, UT Austin**
- Graded assignments and held office hours for a class of 40+ students
 - Shared resources and information regarding ethical and social issues in computer science
- Spring 2019, Fall 2019 **Undergraduate TA, CS361 Introduction to Computer Security, UT Austin**
- Created and graded security-focused assignments for 80+ students
 - Lectured on various topics in security including cryptography and data forensics
 - Wrote, hosted and ran a CTF competition for the students’ final exam

FELLOWSHIPS & HONORS

- 2024 **Doctoral Excellence in Service & Leadership Award**, *UCSD Computer Science & Engineering*
- 2024-2025 **Qualcomm Innovation Fellowship**, *Qualcomm*
- 2021-2025 **San Diego Fellowship**, *UCSD Graduate Division*

CTF AWARDS

- 2020 **AtlassianCTF (1st)**, with team "hhh_"
- 2019 **SunshineCTF (1st)**, **SwampCTF (3rd)**, **AngstromCTF (3rd)**, with team "UTC"
Texas Network Massacre (1st), **AtlassianCTF (3rd)**, with team "hhh_"
- 2018 **AtlassianCTF (3rd)**, with team "hhh_"

LEADERSHIP

- 2023-Present **(Re-)Founder, Graduate Student Lounge Rep.**, *UCSD CSE Graduate Student Council*
 - Led initiative to revive CSE Graduate Student Council, which had dissolved 10+ years prior
- 2022-Present **Administrator**, *UCSD CSE "Chez Bob"*
 - Oversee operation of student-run snack and food co-op that handles hundreds of transactions per day
 - Order and restock food and drinks, maintain cold brew kegerator
 - Maintain infrastructure that runs POS system, fixing bugs and adding features in JavaScript and SQL
- 2018–2021 **President (previously Engineering Officer)**, *UT Information & Systems Security Society*
 - Led the UTCTF project in 2021 and 2020, a CTF with over 2500+ participants. Coordinated planning, communication, prizes, and wrote challenges ([issss.io/github/UTCTF-21](https://github.com/UTCTF-21), [issss.io/github/UTCTF-20](https://github.com/UTCTF-20))
 - Created ForeverCTF, an always-up entry level CTF to allow people to practice security skills (forever.issss.io)
 - Led "Beginner Series" initiative, a series of introductory security talks (issss.io/talks/beginner-series)
 - Wrote security challenges for biweekly Capture the Flag (CTF) competitions (issss.io/github/ctf)
 - Gave talks about security-related topics such as cryptography, data forensics, privacy, etc. (issss.io/talks)
- 2019–2021 **Captain**, *UT Collegiate Cyber Defense Comp. and Collegiate Penetration Testing Comp.*
 - CCDC: Led a team of 8 in a blue team simulation, where students must defend 8-10 machines from red team attackers while also completing business 'injects' (setting up new services, managing users, etc.). Competed at Nationals in 2021, placed 1st (2021), 2nd (2019), 3rd (2020) at Southwest Regionals
 - CPTC: Led a team of 6 students in a red team simulation, where students perform a comprehensive penetration test of a company network with , then write a detailed report of the vulnerabilities and security flaws they found. Placed 2nd (2019) at New England Regionals
- 2018–2020 **Web/Tech Senior Officer (previously Web/Tech Junior Officer)**, *UT ACM Chapter*
 - Implemented new features and fixed bugs on UT's ACM chapter website
 - Wrote curriculum for and hosted 'CS101', a series of 8-10 introductory workshops for freshmen with topics like Linux basics, Git/VCS, debugging, etc (github.com/UTACM/CS101)
 - Created and implemented 'A to Zs of UTCS', a glossary of terms related to computer science, UTCS and UT Austin to help new students get up to speed (texasacm.org/AtoZ)

ADDITIONAL SERVICE

- 2026 **Embedded Area Lead**, *UCSD CSE PhD Visit Day*
- 2022-2024 **GPSA Representative**, *UCSD Student Transit Advisory Council*
- 2022-2024 **CSE Dept. Representative**, *UCSD Graduate & Professional Student Association (GPSA)*

TECHNICAL SKILLS

Most comfortable in Python, C/C++; familiar with Rust, assembly (x86), bash scripting, SQL, JavaScript, HTML/CSS

Comfortable with Linux, git, L^AT_EX, Ghidra, Wireshark, scapy, GIMP, and common command line tools. Familiar with gdb, Frida, LLVM, GNURadio, and Docker.

Comfortable programming using BLE and WiFi radios (nRF52, ESP32), and with underlying protocols (especially BLE, WPA/WPA2 authentication). Experience with soldering, schematic design, PCB layout, using logic analyzers (Saleae), programmers, SDRs. Comfortable with power and hand tools, experience with metal lathe, CNC mill.

Love to reverse and gather device/protocol information using datasheets, patents, specifications/standards, fccid.io, WiGLE, and the general internet.